Chemistry - Graduate Level Assessment 21 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The executioner in charge of the lethal gas chamber at the state penitentiary adds excess dilute H_2 SO_4 to 196 g (about (1/2) lb) ofNaCN. What volume of HCN gas is formed at STP?

   (a) 11.2 liters

   (b) 67.2 liters

   (c) 168 liters

   (d) 100.8 liters

   (e) 22.4 liters

2. Calculate the average separation of the electron and the nucleus in the ground state of the hydrogen atom.

   (a) 0.159 pm

   (b) 79.5 pm

   (c) 210 pm

   (d) 132 pm

   (e) 0.53 pm

3. Calculate the number of $H^{+} ionsintheinnercompartmentofasingleratlivermitochondrion, assuming$

   4. 1000 $H^{+} ions$

   4. 900 $H^{+} ions$

   4. 843 $H^{+} ions$

   4. 950 $H^{+} ions$

   4. 615 $H^{+} ions$

Which is correct about the calcium atom?

(a) It contains 20 protons and neutrons

(b) All atoms of calcium have 20 protons and 40 neutrons

(c) It contains 20 protons, neutrons, and electrons

(d) All calcium atoms have 20 protons, neutrons, and electrons
(e) All atoms of calcium have a mass of 40.078 u

In which of the following is not the negative end of the bond written last?

(a) H-O
(b) P-Cl
(c) N-H
(d) S-O
(e) C-O

True / False (10 marks)

Answer True or False

6. True or False: Gold (Au) is expected to have a higher electronegativity than iron (Fe) because it is a noble metal.

7. True or False: The volume of a rectangular tank measuring 2.0 m by 50 cm by 200 mm is 450 liters.

8. True or False: CBr 4 exhibits a tetrahedral molecular geometry due to its four equivalent bromine atoms surrounding a central carbon atom.

9. True or False: The conjugate acid of the H_2PO_4- ion is H_3PO_4.

10. True or False: The frequency of light emitted by a helium-neon laser is approximately 4.200 x $10^1$4 Hz.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the mass of Ca(OH) required to neutralize 28.0 g of HCl, and discuss how this amount could neutralize phosphoric acid, providing detailed calculations and reasoning.

12. Discuss the relationship between the chemical potentials of water and ethanol in a mixture with a mole fraction of water at 0.60, given an increase in the chemical potential of water by 0.25 J mol^1. How does this change affect the chemical potential of ethanol? Provide a detailed analysis.

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